

Personal Fitness Data Analysis

Hans Dano

2026-08-30

```
#Read Data
```

```
hans_exercise_2025 <- read.csv("hans_dano_exercise_data_2025_to_2026.csv")
```

```
#Load and Explore / Data Cleaning
```

```
head(hans_exercise_2025)
```

```
##      Date Phase muscle_group      Exercise Weight..lbs. Set.Number Reps
## 1 2025-01-09 Bulk        Legs    Hack Squats        270         1     12
## 2 2025-01-09 Bulk        Legs    Hack Squats        270         2     10
## 3 2025-01-09 Bulk        Legs    Hack Squats        270         3      8
## 4 2025-01-09 Bulk        Legs Seated Leg Curls    160         1     10
## 5 2025-01-09 Bulk        Legs Seated Leg Curls    160         2     10
## 6 2025-01-09 Bulk        Legs Seated Leg Curls    160         3      8
##      Bodyweight      Notes.
## 1      NA do not account for machine weight
## 2      NA do not account for machine weight
## 3      NA do not account for machine weight
## 4      NA
## 5      NA
## 6      NA
```

```
str(hans_exercise_2025)
```

```
## 'data.frame':    1066 obs. of  9 variables:
## $ Date      : chr  "2025-01-09" "2025-01-09" "2025-01-09" "2025-01-09" ...
## $ Phase     : chr  "Bulk" "Bulk" "Bulk" "Bulk" ...
## $ muscle_group: chr  "Legs" "Legs" "Legs" "Legs" ...
## $ Exercise  : chr  "Hack Squats" "Hack Squats" "Hack Squats" "Seated Leg Curls" ...
## $ Weight..lbs.: num  270 270 270 160 160 160 180 180 220 220 ...
## $ Set.Number : int   1 2 3 1 2 3 1 2 1 2 ...
## $ Reps      : num   12 10 8 10 10 8 10 10 12 10 ...
## $ Bodyweight : num   NA NA NA NA NA NA NA NA NA NA ...
## $ Notes.    : chr  "do not account for machine weight" "do not account for machine weight" "do not account for machine weight" ...
```

```
summary(hans_exercise_2025)
```

```
##      Date      Phase      muscle_group      Exercise
```

```
## Length:1066      Length:1066      Length:1066      Length:1066
## Class :character  Class :character  Class :character  Class :character
## Mode :character   Mode :character   Mode :character   Mode :character
##
##
##
##
## Weight..lbs.      Set.Number      Reps      Bodyweight
## Min. : 0.0        Min. :1.000    Min. : 3.000  Min. :135.1
## 1st Qu.: 42.5      1st Qu.:1.000    1st Qu.: 7.000  1st Qu.:138.0
## Median :100.0      Median :2.000    Median : 8.000  Median :148.8
## Mean :106.7        Mean :1.841     Mean : 8.201    Mean :146.7
## 3rd Qu.:150.0      3rd Qu.:2.000    3rd Qu.: 8.000  3rd Qu.:152.6
## Max. :370.0        Max. :4.000     Max. :30.000    Max. :158.0
##
##                  NA's :3      NA's :709
## Notes.
## Length:1066
## Class :character
## Mode :character
##
##
##
##
```

#Check all column names

```
colnames(hans_exercise_2025)
```

```
## [1] "Date"      "Phase"      "muscle_group" "Exercise"    "Weight..lbs."
## [6] "Set.Number" "Reps"      "Bodyweight"  "Notes."
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.3
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

#Rename columns

```
#Weight..lbs. to Weight(lbs)
```

```
hans_exercise_2025 <- hans_exercise_2025 %>%
```

```
  rename(Weight_lbs = 5) #Weight..lbs. doesnt work so instead enter the index number "5"
```

```
#Set.Number to set_number
hans_exercise_2025 <- hans_exercise_2025 %>%
  rename(set_number = Set.Number)
```

```
#Notes. to Notes
hans_exercise_2025 <- hans_exercise_2025 %>%
  rename(Notes = Notes.)
```

```
#Standardize all column names to lowercase
colnames(hans_exercise_2025) <- tolower(colnames(hans_exercise_2025))
```

#Automatically added two extra columns when exporting so delete them

```
#hans_exercise_2025 <- hans_exercise_2025 %>% select(-c(x, x.1))
```

```
# Using any_of() allows the code to still run if those columns are no longer present.
hans_exercise_2025 <- hans_exercise_2025 %>%
  select(-any_of(c("x", "x.1")))
```

#Check if success

```
colnames(hans_exercise_2025)
```

```
## [1] "date"      "phase"      "muscle_group" "exercise"    "weight_lbs"
## [6] "set_number" "reps"       "bodyweight"  "notes"
```

#Check the different row names for each column where it makes sense (Phase, muscle_group, Exercise)

```
hans_exercise_2025 %>%
  distinct(phase)
```

```
##   phase
## 1 Bulk
## 2 Cut
```

```
#or unique(hans_exercise_2025$phase)
```

```
hans_exercise_2025 %>%
  distinct(muscle_group)
```

```
##   muscle_group
## 1      Legs
## 2      Legs
## 3      Push
## 4      Pull
## 5      Push
```

```
#Output:
```

```
#Legs
```

```
#Legs
```

```
#Push
```

```
#Pull
```

```
#Push
```

```
#Obviously this is wrong since it created multiple distinct when there are just 3
```

```
#Find a fix to the above: try turning everything to lower case, did not work
```

```
hans_exercise_2025 %>%
```

```
  mutate(muscle_group = tolower(muscle_group)) %>%
```

```
  distinct(muscle_group)
```

```
##   muscle_group
```

```
## 1         legs
```

```
## 2         legs
```

```
## 3         push
```

```
## 4         pull
```

```
## 5         push
```

```
library(stringr)
```

```
## Warning: package 'stringr' was built under R version 4.5.3
```

```
hans_exercise_2025 %>%
```

```
  mutate(muscle_group = str_trim(muscle_group), # cuts invisible leading and trailing spaces.
```

```
         muscle_group = str_squish(muscle_group), #removes repeated spaces inside the text
```

```
         muscle_group = str_to_title(muscle_group)) %>% # forces everything to standard title case (so
```

```
  distinct(muscle_group)
```

```
##   muscle_group
```

```
## 1         Legs
```

```
## 2         Push
```

```
## 3         Pull
```

```
#Output: Legs, Push, Pull
```

```
#Save this change:
```

```
hans_exercise_2025 <- hans_exercise_2025 %>%
```

```
  mutate(muscle_group = str_trim(muscle_group),
```

```
         muscle_group = str_squish(muscle_group),
```

```
         muscle_group = str_to_title(muscle_group))
```

```
#Fix column value names
```

```
hans_exercise_2025 %>%
```

```
  distinct(exercise) %>%
```

```
  arrange(exercise) #sort alphabetically
```

```

##                exercise
## 1              Back Extensions
## 2              Bar Cable Curls
## 3              Bar Overhead Tricep
## 4              Bar Pushdown Tricep
## 5              Barbell RDL
## 6              Barbell Rows
## 7              Bayesian Curls
## 8              Bench Press
## 9              Bulgarian Split Squats
## 10             Cable Lateral Raise
## 11             Cable Pullovers
## 12             Chest Fly
## 13 Diverging Isolated Lat Pulldown
## 14             Dumbell Lateral Raises
## 15             Dumbell Rear Delt
## 16             EZ Bar Overhead Tricep
## 17             EZ Bar Preacher Curl
## 18             Hack Squats
## 19             Hip Thrust
## 20             Incline Bench Press
## 21             Incline Dumbell Press
## 22             Isolated Lat Pull Down
## 23             Isolated Overhead Tricep
## 24             Isolated Pushdown Tricep
## 25             Isolated Rear Delt Cable
## 26 Isolated Reverse Pec Dec Rear Delts
## 27             Isolated Seated Machine Rows
## 28             Kickback Tricep
## 29 Kneeling Isolated Lat Pulldown
## 30             Lat Pulldown
## 31             Lat Pulldown (Cybex)
## 32             Leg Extensions
## 33             Leg Extenstions
## 34             Leg Press
## 35             Leg Press (Precor)
## 36             Lying Leg Curls
## 37             Machine Lateral Raise
## 38             Machine Shoulder Press
## 39             Military Press
## 40             Pec Dec
## 41             Pullups
## 42             Pullups
## 43             Rear Delt Cable
## 44             Revers Pec Dec Rear Delts
## 45             Reverse Pec Dec Rear Delts
## 46             Rope Front Delt Raises
## 47             Rope Hammer Curls
## 48             Rope Overhead Tricep
## 49             Rope Pushdown Tricep
## 50             Rope Rear Delts
## 51             Seated Cable Chest Press
## 52             Seated Cable Rows
## 53             Seated Cable Rows

```

```
## 54      Seated Cable Rows (Precor)
## 55      Seated Calf Raises
## 56      Seated Dumbbell Curls
## 57      Seated Dumbbell Hammer Curls
## 58      Seated Leg Curls
## 59      Seated Machine Incline Press
## 60      Seated Machine Preacher Curl
## 61      Seated Machine Rows
## 62      Seated Machine Rows (Cybex)
## 63      Seated Military Press
## 64      Smith Machine Incline Press
## 65      Smith Machine JM Press
## 66      Standing Calf Raises
## 67      Standing Dumbbell Curls
## 68      Standing Dumbbell Hammer Curls
## 69      T Bar Rows
## 70      Tricep Kickback
## 71      Upward Chest Fly
## 72      Weighted Dips
## 73 isolated Reverse Pec Dec Rear Delts
```

```
#Found "Leg Extensions" and "Leg Extentions"
#Found "Revers Pec Dec Rear Delts" and Reverse "Pec Dec Rear Delts"
#Found 2 copies of Seated Cable Rows (not including Precor)
```

```
#Fix 1: Found "Leg Extensions" and "Leg Extentions"
hans_exercise_2025 <- hans_exercise_2025 %>%
  mutate(exercise = case_match(exercise, "Leg Extentions" ~ "Leg Extensions", .default = exercise))
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'exercise = case_match(exercise, "Leg Extentions" ~ "Leg
##   Extensions", .default = exercise)'.
## Caused by warning:
## ! 'case_match()' was deprecated in dplyr 1.2.0.
## i Please use 'recode_values()' instead.
```

```
#Fix 2: Found "Revers Pec Dec Rear Delts" and Reverse "Pec Dec Rear Delts"
hans_exercise_2025 <- hans_exercise_2025 %>%
  mutate(exercise = case_match(exercise, "Revers Pec Dec Rear Delts" ~ "Reverse Pec Dec Rear Delts", .d
```

```
#Fix 3: Found 2 copies of Seated Cable Rows (not including Precor)
hans_exercise_2025 <- hans_exercise_2025 %>%
  mutate(exercise = ifelse(str_detect(exercise, "Seated Cable Rows") & !str_detect(exercise, "Precor"),
#Condition: If value contains "Seated Cable Rows" and does NOT contain "Precor", then we want it to be
```

```
#I also noticed there's 1127 observations but only 1066 in csv file
```

```
#Check the 1067th row:
hans_exercise_2025 %>%
  slice(1067) #OR hans_exercise_2025[1068, ] where [rownum, colnum]
```

```
## [1] date      phase      muscle_group exercise  weight_lbs
## [6] set_number reps      bodyweight notes
## <0 rows> (or 0-length row.names)
```

```
#First check the first 1066 rows and see if it contains BLANK in ALL the column values
blank_data <- which(hans_exercise_2025$date == "" & hans_exercise_2025$phase == "" & hans_exercise_2025$
blank_data
```

```
## integer(0)
```

```
#Drop the last 1127 - 1066 = 61 rows
```

```
#So rows 1067 to 1127 are empty. We can then delete them
nrow(hans_exercise_2025[-c(1067:1127), ])
```

```
## [1] 1066
```

```
hans_exercise_2025 <- hans_exercise_2025[-c(1067:1127), ]
```

```
hans_exercise_2025 %>%
  distinct(date)
```

```
##      date
## 1 2025-01-09
## 2 2025-01-10
## 3 2025-01-12
## 4 2025-01-13
## 5 2025-01-15
## 6 2025-01-16
## 7 2025-01-17
## 8 2025-02-05
## 9 2025-02-06
## 10 2025-02-08
## 11 2025-02-10
## 12 2025-02-11
## 13 2025-02-14
## 14 2025-02-19
## 15 2025-02-22
## 16 2025-02-24
## 17 2025-02-25
## 18 2025-02-26
## 19 2025-02-27
## 20 2025-02-28
## 21 2025-03-03
## 22 2025-03-04
## 23 2025-03-05
## 24 2025-03-07
## 25 2025-03-17
## 26 2025-03-18
## 27 2025-03-19
## 28 2025-03-20
## 29 2025-03-24
## 30 2025-03-25
## 31 2025-03-26
```

32 2025-03-27
33 2025-03-28
34 2025-03-31
35 2025-04-01
36 2025-04-02
37 2025-04-03
38 2025-04-07
39 2025-04-08
40 2025-04-09
41 2025-04-10
42 2025-04-17
43 2025-05-20
44 2025-06-05
45 2025-06-09
46 2025-06-11
47 2025-06-12
48 2025-06-16
49 2025-06-17
50 2025-06-18
51 2025-06-23
52 2025-06-24
53 2025-06-25
54 2025-06-26
55 2025-06-27
56 2025-06-30
57 2025-07-02
58 2025-07-03
59 2025-07-07
60 2025-07-08
61 2025-07-14
62 2025-07-15
63 2025-07-17
64 2025-07-21
65 2025-07-28
66 2025-07-29
67 2025-07-31
68 2025-08-05
69 2025-08-11
70 2025-08-12
71 2025-08-14
72 2025-08-16
73 2025-08-18
74 2025-08-21
75 2025-08-22
76 2025-08-25
77 2025-08-26
78 2025-09-02
79 2025-09-05
80 2025-09-08
81 2025-09-11
82 2025-09-22
83 2025-09-23
84 2025-10-02
85 2025-10-06

```
## 86 2025-10-07
## 87 2025-10-10
## 88 2025-10-14
## 89 2025-10-15
## 90 2025-10-16
## 91 2025-10-17
## 92 2025-10-22
## 93 2025-10-23
## 94 2025-10-27
## 95 2025-10-28
```

```
hans_exercise_2025 %>% #Need to make 0.0 value be NA since this would skew our data
  distinct(weight_lbs)
```

```
##      weight_lbs
## 1          270.0
## 2          160.0
## 3          180.0
## 4          220.0
## 5          205.0
## 6           45.0
## 7          225.0
## 8           80.0
## 9           90.0
## 10           0.0
## 11          55.0
## 12          70.0
## 13          35.0
## 14          40.0
## 15          110.0
## 16          100.0
## 17           87.5
## 18          140.0
## 19           50.0
## 20          320.0
## 21          200.0
## 22          170.0
## 23          120.0
## 24          185.0
## 25           42.5
## 26           30.0
## 27          105.0
## 28          115.0
## 29          240.0
## 30           32.5
## 31           25.0
## 32          175.0
## 33          165.0
## 34          150.0
## 35          130.0
## 36          360.0
## 37          370.0
## 38          210.0
## 39          145.0
```

```
## 40      190.0
## 41      230.0
## 42      245.0
## 43       92.5
## 44       95.0
## 45       27.5
## 46       60.0
## 47      290.0
## 48      135.0
## 49      155.0
## 50      125.0
## 51       37.5
```

```
hans_exercise_2025 %>%
  distinct(set_number)
```

```
##   set_number
## 1           1
## 2           2
## 3           3
## 4           4
```

```
hans_exercise_2025 %>% #There shouldn't be NA reps so remove columns where it says that
  distinct(reps)
```

```
##   reps
## 1 12.0
## 2 10.0
## 3  8.0
## 4  4.0
## 5  3.0
## 6  7.5
## 7  6.0
## 8  9.0
## 9  7.0
## 10 NA
## 11 18.0
## 12 15.0
## 13  8.5
## 14 14.0
## 15 11.0
## 16  5.0
## 17 20.0
## 18 30.0
## 19 17.0
## 20  6.5
## 21 22.0
```

```
hans_exercise_2025 %>%
  distinct(bodyweight)
```

```
##   bodyweight
```

```
## 1      NA
## 2      158.0
## 3      155.2
## 4      154.8
## 5      154.6
## 6      153.8
## 7      153.4
## 8      152.4
## 9      152.6
## 10     152.0
## 11     150.2
## 12     150.0
## 13     149.4
## 14     148.4
## 15     149.0
## 16     148.8
## 17     146.0
## 18     147.0
## 19     145.2
## 20     144.0
## 21     141.8
## 22     138.2
## 23     138.0
## 24     137.5
## 25     135.8
## 26     136.9
## 27     136.0
## 28     135.4
## 29     135.1
```

```
hans_exercise_2025 %>%
  distinct(notes)
```

```
##              notes
## 1 do not account for machine weight
## 2
## 3 misgrooved twice
## 4 bodyweight is NA
## 5 didn't track reps went to failure
## 6 Alternating
## 7 Weight in Each hand
## 8 Precor
## 9 Felt something in my lower back
## 10 Cross non isolated
## 11 It was a battle
## 12 Isolated
## 13 Neutral grip, weight kept slipping
## 14 weight on each side
## 15 lower focused
## 16 Cybex
## 17 Only 2 sets since chronic left glute soreness
## 18 I think theres a stability issue. I struggled with a 5 lb difference
## 19 Neutral Grip
## 20 Wide grip
```

```
## 21                                No Weight
## 22                                Did not push myself
## 23                                Strict new form
## 24                                felt really weak here
## 25                                PR!
## 26                                Cybex, underhand
## 27                                Non Isolated
## 28                                More strict
## 29                                Felt Weak today
## 30                                pr!
## 31                                Feel like deloaded today
## 32                                focus on strict form today
## 33                                rough 8
## 34                                lower focus
## 35                                Did not rest long enough
## 36                                Didn't rest long enough
## 37                                STRICT
## 38                                Got sick but we back
```

```
#Change all 0.0 values (wherever they are) in weight_lbs to be NA
which(is.na(hans_exercise_2025$reps) | hans_exercise_2025$reps < 1) ## 0.0 does not work for doubles
```

```
## [1] 28 29 30
```

```
#OR
hans_exercise_2025 %>%
  filter(reps < 1 | is.na(reps))
```

```
##      date phase muscle_group      exercise weight_lbs set_number reps
## 1 2025-01-12 Bulk      Pull Rope Rear Delts        80         1    NA
## 2 2025-01-12 Bulk      Pull Rope Rear Delts        80         2    NA
## 3 2025-01-12 Bulk      Pull Rope Rear Delts        80         3    NA
##  bodyweight      notes
## 1      NA didn't track reps went to failure
## 2      NA didn't track reps went to failure
## 3      NA didn't track reps went to failure
```

```
# So the notes actually say I didn't track reps so this isn't useful at all. I'm just going to delete i
hans_exercise_2025 <- hans_exercise_2025 %>%
  filter(reps >= 1 | !is.na(reps))
```

```
#Convert date to an actual date format
```

```
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 4.5.3
```

```
##
```

```
## Attaching package: 'lubridate'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      date, intersect, setdiff, union
```

```
hans_exercise_2025 <- hans_exercise_2025 %>%
  mutate(date = ymd(date))
```

#FEATURE ENGINEERING

```
#Add a new column called volume, which calculates weight * reps
hans_exercise_2025 <- hans_exercise_2025 %>%
  mutate(volume = weight_lbs * reps)
```

#Epley Formula

```
#Did some research and we can actually implement what's called the "Epley Formula" which calculates a p

#1RM = w * (1 + r/30)
#w is weight lifted
#r is number of successful reps

#Note: this will only be useful for lifts that are between 2 to 10 repetitions since any higher than th

hans_exercise_2025 <- hans_exercise_2025 %>%
  mutate(one_rep_max = weight_lbs * (1 + reps/30))
```

#Make 2 different trimmed data frames that are strictly bulk phase and cutting phase

```
bulk_hans_exercise_2025 <- hans_exercise_2025 %>%
  filter(phase == "Bulk")
```

```
cut_hans_exercise_2025 <- hans_exercise_2025 %>%
  filter(phase == "Cut")
```

#We can now group by the data for cut and bulk into weeks. This way, analyzing can be easier

```
#Step 1: We can use function floor_date which adds a column that rounds the time unit DOWN to the nearest
bulk_hans_exercise_2025 <- bulk_hans_exercise_2025 %>%
  mutate(start_of_week = floor_date(date, unit = "week", week_start = 1))

#Check what this looks like
head(bulk_hans_exercise_2025)
```

##	date	phase	muscle_group	exercise	weight_lbs	set_number	reps
## 1	2025-01-09	Bulk	Legs	Hack Squats	270	1	12
## 2	2025-01-09	Bulk	Legs	Hack Squats	270	2	10
## 3	2025-01-09	Bulk	Legs	Hack Squats	270	3	8
## 4	2025-01-09	Bulk	Legs	Seated Leg Curls	160	1	10
## 5	2025-01-09	Bulk	Legs	Seated Leg Curls	160	2	10
## 6	2025-01-09	Bulk	Legs	Seated Leg Curls	160	3	8
##	bodyweight	notes	volume	one_rep_max	start_of_week		
## 1	NA	do not account for machine weight	3240	378.0000	2025-01-06		
## 2	NA	do not account for machine weight	2700	360.0000	2025-01-06		
## 3	NA	do not account for machine weight	2160	342.0000	2025-01-06		
## 4	NA		1600	213.3333	2025-01-06		
## 5	NA		1600	213.3333	2025-01-06		
## 6	NA		1280	202.6667	2025-01-06		

#Now we can group by into the unique buckets: start of week, muscle group, exercise #On top of this, we will aggregate using summarise() to give us weekly sets per exercise, weekly max_weight per exercise, weekly max 1RM, weekly total volume (average_body_weight will only work for cut_weekly data)

```
#weekly_sets will count how many rows (sets) per week we had
#max_weight will look at all the sets in the bucket, and will look at the highest weight
#max_1_rm will take the highest one rep max considering each bucket
#total_volume will scan the volume in all the rows per bucket then sum it
#average_body_weight will take all the body weights per bucket then average it out
bulk_weekly <- bulk_hans_exercise_2025 %>%
  group_by(start_of_week, muscle_group, exercise) %>%
  summarise(weekly_sets = n(),
            max_weight = max(weight_lbs, na.rm = TRUE), #have to remove NAs or it compiles blank
            max_1_rm = max(one_rep_max, na.rm = TRUE),
            total_volume = sum(volume, na.rm = TRUE),
            average_bodyweight = mean(bodyweight, na.rm = TRUE),
            .groups = "drop"
  )

bulk_weekly
```

```
## # A tibble: 164 x 8
##   start_of_week muscle_group exercise      weekly_sets max_weight max_1_rm
##   <date>         <chr>      <chr>          <int>      <dbl>    <dbl>
## 1 2025-01-06     Legs      Hack Squats          3        270      378
## 2 2025-01-06     Legs      Leg Extensions        2        180      240
## 3 2025-01-06     Legs      Seated Leg Curls       3        160      213.
## 4 2025-01-06     Legs      Standing Calf Rai~     3        220      308
## 5 2025-01-06     Pull      Barbell Rows          5        225      285
## 6 2025-01-06     Pull      Isolated Seated M~     2         90      114
## 7 2025-01-06     Pull      Pullups               3         0         0
## 8 2025-01-06     Push      Bench Press           3        205      260.
## 9 2025-01-06     Push      Rope Pushdown Tri~     3         90      114
## 10 2025-01-06     Push      Weighted Dips         3         45       57
## # i 154 more rows
## # i 2 more variables: total_volume <dbl>, average_bodyweight <dbl>
```

#Do the same for cut

```
cut_hans_exercise_2025 <- cut_hans_exercise_2025 %>%
  mutate(start_of_week = floor_date(date, unit = "week", week_start = 1))

#Check what this looks like
head(cut_hans_exercise_2025)
```

```
##           date phase muscle_group      exercise weight_lbs
## 1 2025-06-09   Cut      Push Smith Machine Incline Press    190
## 2 2025-06-09   Cut      Push Smith Machine Incline Press    190
## 3 2025-06-09   Cut      Push Smith Machine Incline Press    190
## 4 2025-06-09   Cut      Push      Seated Cable Chest Press    150
## 5 2025-06-09   Cut      Push      Seated Cable Chest Press    130
## 6 2025-06-09   Cut      Push      Cable Lateral Raise       40
##   set_number reps bodyweight notes volume one_rep_max start_of_week
```

```
## 1      1      7      158      1330      234.33333      2025-06-09
## 2      2      7      158      1330      234.33333      2025-06-09
## 3      3      6      158      1140      228.00000      2025-06-09
## 4      1      7      158      1050      185.00000      2025-06-09
## 5      2      7      158      910      160.33333      2025-06-09
## 6      1      8      158      320      50.66667      2025-06-09
```

```
cut_weekly <- cut_hans_exercise_2025 %>%
  group_by(start_of_week, muscle_group, exercise) %>%
  summarise(weekly_sets = n(),
            max_weight = max(weight_lbs, na.rm = TRUE), #have to remove NAs or it compiles blank
            max_1_rm = max(one_rep_max, na.rm = TRUE),
            total_volume = sum(volume, na.rm = TRUE),
            average_bodyweight = mean(bodyweight, na.rm = TRUE),
            .groups = "drop"
  )

cut_weekly
```

```
## # A tibble: 218 x 8
##   start_of_week muscle_group exercise      weekly_sets max_weight max_1_rm
##   <date>         <chr>      <chr>          <int>      <dbl>    <dbl>
## 1 2025-06-09    Pull      Lat Pulldown         3        170    215.
## 2 2025-06-09    Pull      Seated Cable Rows     3        140    177.
## 3 2025-06-09    Pull      Standing Dumbbell ~     1         40    50.7
## 4 2025-06-09    Push      Bar Cable Curls        3        110    139.
## 5 2025-06-09    Push      Bar Pushdown Tric~     3        110    139.
## 6 2025-06-09    Push      Cable Lateral Rai~     9         45    55.5
## 7 2025-06-09    Push      Rope Front Delt R~     3         60     76
## 8 2025-06-09    Push      Rope Overhead Tri~     5        110    139.
## 9 2025-06-09    Push      Rope Pushdown Tri~     3        120    152
## 10 2025-06-09   Push      Seated Cable Ches~     2        150    185
## # i 208 more rows
## # i 2 more variables: total_volume <dbl>, average_bodyweight <dbl>
```

#Question 1: Does my weekly training volume predict my weekly weight loss during my cut, after controlling for exercise type?

#To answer this, we don't need 1 row per exercise. Exercise is irrelevant here and so we need to group

```
cut_weekly_total <- cut_hans_exercise_2025 %>%
  group_by(start_of_week) %>%
  summarise(total_weekly_volume = sum(volume, na.rm = TRUE), #total volume for each week
            avg_bodyweight = mean(bodyweight, na.rm = TRUE), #average bodyweight for each week
            .groups = "drop"
  ) %>%
  mutate(weight_change = avg_bodyweight - lag(avg_bodyweight)) #add a new column for weight change

cut_weekly_total
```

```
## # A tibble: 20 x 4
##   start_of_week total_weekly_volume avg_bodyweight weight_change
```

	<date>	<dbl>	<dbl>	<dbl>
## 1	2025-06-09	28432.	156.	NA
## 2	2025-06-16	48915	155.	-1.66
## 3	2025-06-23	57388.	153.	-1.52
## 4	2025-06-30	26180	152.	-0.818
## 5	2025-07-07	11195	150.	-2.26
## 6	2025-07-14	22930	149.	-1.16
## 7	2025-07-21	7020	NaN	NaN
## 8	2025-07-28	22645	147.	NaN
## 9	2025-08-04	7420	147	-0.107
## 10	2025-08-11	22388.	144.	-2.61
## 11	2025-08-18	19185	NaN	NaN
## 12	2025-08-25	11748.	142.	NaN
## 13	2025-09-01	16160	NaN	NaN
## 14	2025-09-08	14085	NaN	NaN
## 15	2025-09-22	14050	138.	NaN
## 16	2025-09-29	10855	138	-0.200
## 17	2025-10-06	16390	136.	-1.69
## 18	2025-10-13	29115	136.	-0.0910
## 19	2025-10-20	14210	136	-0.219
## 20	2025-10-27	16870	135.	-0.738

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

#Now we will visualize the relationship between the our independent variable: weekly volume vs. dependent variable: weight change

```
ggplot(data = cut_weekly_total, aes(x = total_weekly_volume, y = weight_change)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) + #method = "lm" tells ggplot2 to make a best of fit LINEAR line
  labs(
    title = "Weekly Volume vs Weight Change (Cut)",
    x = "Total Weekly Volume (lbs)",
    y = "Weight Change (lbs) - Negative = Loss"
  )
```

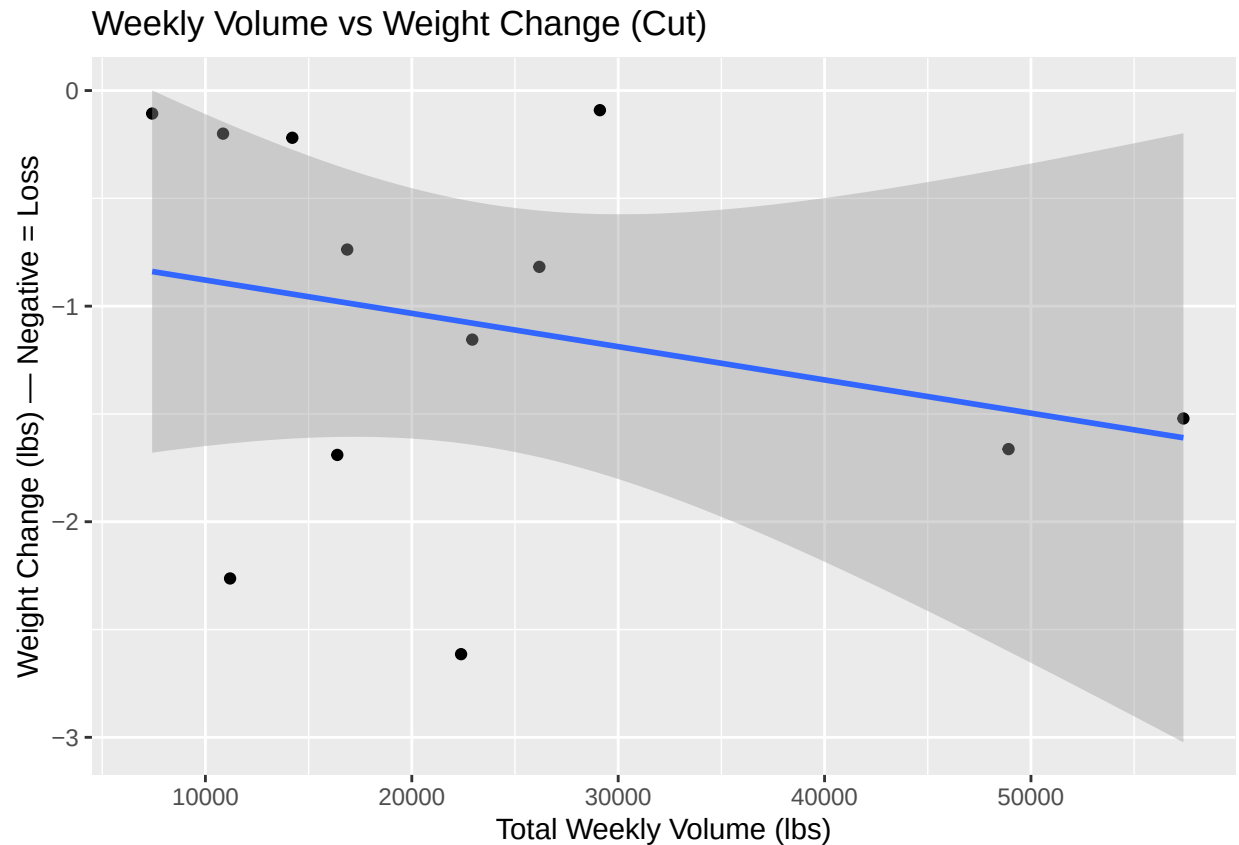
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
```

```
## ('geom_point()').
```



#Although the plots seem random, we will do a hypothesis test considering our slope, p-value, and coefficient of determination (r^2)

```
#Observe the intercept and slope
cut_model <- lm(weight_change ~ total_weekly_volume, data = cut_weekly_total)
cut_model
```

```
##
## Call:
## lm(formula = weight_change ~ total_weekly_volume, data = cut_weekly_total)
##
## Coefficients:
##      (Intercept)  total_weekly_volume
##      -7.248e-01    -1.543e-05
```

```
#Slope: For every 1 lb of volume, weight_change changes by -1.541e-05.
```

```
summary(cut_model)
```

```
##
## Call:
## lm(formula = weight_change ~ total_weekly_volume, data = cut_weekly_total)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -1.5439 -0.3155  0.1686  0.7005  1.0832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -7.248e-01  4.797e-01  -1.511    0.162
## total_weekly_volume -1.543e-05  1.723e-05  -0.896    0.392
##
## Residual standard error: 0.8762 on 10 degrees of freedom
## (8 observations deleted due to missingness)
## Multiple R-squared:  0.07425,    Adjusted R-squared:  -0.01832
## F-statistic: 0.8021 on 1 and 10 DF,  p-value: 0.3915
```

#P-value interpretation: If the null hypothesis is true: weekly volume does NOT predict weight change, there is a 39.2% probability we would see a test statistic (the slope) as extreme as the one we observed which is -1.541e-05 or -0.00001541

#Conclusion: So we can say that because p-value = 0.3923 > 0.05, there is no significant statistical evidence and we fail to reject the null hypothesis: “weekly volume does not predict weight change”. The weak negative trend in the data was likely due to random chance.

#R² is how much the percentage of Y variance is explained by X. We can see adjusted R² = -0.01857 is negative (R² more useful than just R² = 0.07403 since it handles multiple predictors) meaning the model is poor. So with this data, it is interpreted as -1.86% of Y’s variance is explained by X which doesn’t make a lot of sense and effectively means none of the variance in weight loss is explained by weekly_volume. This must mean that 12 weeks is just too small of a samples size

#After searching, this is the formula for R² and how it was used in my dataset: #Adjusted R² = 1 - [(1 - R²) × (n - 1) / (n - k - 1)] where k is number of predictors # = 1 - [(1 - 0.074) × (12 - 1) / (12 - 1 - 1)] # = 1 - [0.926 × 11 / 10] # = 1 - [0.926 × 1.1] # = 1 - 1.0186 # = -0.0186

#Question 2: During my bulk, I wondered if my incline dumbbell press strength increased when I had more rest (Disclaimer: I know that some days I just forgot to record which can be mistaken as a rest day and could skew the data. To account for this, I want to define a rest day as <= 7 day gap from doing the push muscle group. This is because I did chest without a gap greater than 7 for my split, Otherwise, if the gap is >7 days, we can consider this a logging gap in data)

#First, we will make a new subset that filters for any push exercise (This is because they are all usef

```
push_bulk <- bulk_hans_exercise_2025 %>%
  filter(muscle_group == "Push")
head(push_bulk)
```

```
##      date phase muscle_group      exercise weight_lbs set_number reps
## 1 2025-01-10 Bulk          Push   Bench Press        205         1     8
## 2 2025-01-10 Bulk          Push   Bench Press        205         2     4
## 3 2025-01-10 Bulk          Push   Bench Press        205         3     3
## 4 2025-01-10 Bulk          Push Weighted Dips         45         1     8
## 5 2025-01-10 Bulk          Push Weighted Dips         45         2     8
## 6 2025-01-10 Bulk          Push Weighted Dips         45         3     8
##  bodyweight      notes volume one_rep_max start_of_week
## 1      NA misgrooved twice  1640   259.6667  2025-01-06
## 2      NA misgrooved twice   820   232.3333  2025-01-06
## 3      NA                    615   225.5000  2025-01-06
## 4      NA                    360    57.0000  2025-01-06
```

```
## 5      NA      360      57.0000      2025-01-06
## 6      NA      360      57.0000      2025-01-06
```

```
push_bulk %>%
  distinct(exercise)
```

```
##           exercise
## 1      Bench Press
## 2    Weighted Dips
## 3   Rope Pushdown Tricep
## 4 Smith Machine Incline Press
## 5      Cable Lateral Raise
## 6      Incline Dumbbell Press
## 7   Seated Cable Chest Press
## 8   Isolated Lat Pull Down
## 9   Isolated Pushdown Tricep
## 10  Rope Overhead Tricep
## 11 Standing Dumbbell Hammer Curls
## 12   Isolated Overhead Tricep
## 13   Isolated Rear Delt Cable
## 14 Seated Machine Incline Press
## 15                Chest Fly
## 16   Dumbbell Lateral Raises
## 17                Pec Dec
## 18   Machine Shoulder Press
```

#Now we will calculate the rest days between each push session and clean all the gaps >7days

```
push_bulk <- push_bulk %>%
  arrange(date) %>% #Not necessary to order date since already ordered but safer
  mutate(rest_days_raw = as.numeric(date - lag(date))) %>% #Make a new column of simple current date -
  mutate(rest_days = as.numeric(ifelse(rest_days_raw > 7, "NA", rest_days_raw))) #Make another column w
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'rest_days = as.numeric(ifelse(rest_days_raw > 7, "NA",
##   rest_days_raw))'.
## Caused by warning:
## ! NAs introduced by coercion
```

I also used as.numeric() function around rest_days because it was giving it as chr

```
head(push_bulk)
```

```
##           date phase muscle_group      exercise weight_lbs set_number reps
## 1 2025-01-10 Bulk      Push   Bench Press      205         1      8
## 2 2025-01-10 Bulk      Push   Bench Press      205         2      4
## 3 2025-01-10 Bulk      Push   Bench Press      205         3      3
## 4 2025-01-10 Bulk      Push Weighted Dips      45         1      8
## 5 2025-01-10 Bulk      Push Weighted Dips      45         2      8
## 6 2025-01-10 Bulk      Push Weighted Dips      45         3      8
##   bodyweight      notes volume one_rep_max start_of_week rest_days_raw
## 1      NA misgrooved twice  1640   259.6667   2025-01-06           NA
```

```
## 2      NA misgrooved twice      820      232.3333      2025-01-06      0
## 3      NA                    615      225.5000      2025-01-06      0
## 4      NA                    360       57.0000      2025-01-06      0
## 5      NA                    360       57.0000      2025-01-06      0
## 6      NA                    360       57.0000      2025-01-06      0
## rest_days
## 1      NA
## 2      0
## 3      0
## 4      0
## 5      0
## 6      0
```

```
dim(push_bulk)
```

```
## [1] 231 14
```

```
#Now we filter what we're actually interested in: Incline Dumbell Press.
```

```
#THIS CODE IS BASICALLY FILTERING INCLINE DUMBBELL PRESS BUT TAKING INTO ACCOUNT THE REST GAP FROM THE L
```

```
#We will filter for the specific exercise and take out the rest days that are labelled NA (rest days >
```

```
incline_dumbell_bulk <- push_bulk %>%
```

```
  filter(exercise == "Incline Dumbell Press" & !is.na(rest_days))
```

```
head(incline_dumbell_bulk)
```

```
## [1] date      phase      muscle_group exercise      weight_lbs
## [6] set_number reps      bodyweight  notes        volume
## [11] one_rep_max start_of_week rest_days_raw rest_days
## <0 rows> (or 0-length row.names)
```

```
#came up as an empty data frame.... let's fix this
```

```
push_bulk <- push_bulk %>%
```

```
  mutate(exercise = str_trim(exercise),           # Removes leading/trailing spaces
```

```
         exercise = str_squish(exercise))         # Removes repeated internal spaces
```

```
#Try again
```

```
incline_dumbell_bulk <- push_bulk %>%
```

```
  filter(exercise == "Incline Dumbell Press" & !is.na(rest_days))
```

```
head(incline_dumbell_bulk)
```

```
##      date phase muscle_group      exercise weight_lbs set_number
## 1 2025-01-16 Bulk      Push Incline Dumbell Press      87.5         1
## 2 2025-01-16 Bulk      Push Incline Dumbell Press      87.5         2
## 3 2025-01-16 Bulk      Push Incline Dumbell Press      87.5         3
## 4 2025-02-06 Bulk      Push Incline Dumbell Press      87.5         2
## 5 2025-02-06 Bulk      Push Incline Dumbell Press      87.5         3
## 6 2025-02-10 Bulk      Push Incline Dumbell Press      90.0         1
```

```
##      reps bodyweight      notes volume one_rep_max start_of_week
## 1      8          NA Weight in Each hand 700.0    110.8333 2025-01-13
## 2      7          NA Weight in Each hand 612.5    107.9167 2025-01-13
## 3      6          NA Weight in Each hand 525.0    105.0000 2025-01-13
## 4      8          NA              700.0    110.8333 2025-02-03
## 5      5          NA              437.5    102.0833 2025-02-03
## 6      8          NA              720.0    114.0000 2025-02-10
##      rest_days_raw rest_days
## 1              3         3
## 2              0         0
## 3              0         0
## 4              0         0
## 5              0         0
## 6              2         2
```

```
dim(incline_dumbbell_bulk)
```

```
## [1] 19 14
```

#Fixed!

#For some reason it still shows NA so I just changed the filter from filter(exercise == "Incline Dumbell

```
incline_dumbbell_bulk <- push_bulk %>%
  filter(exercise == "Incline Dumbell Press" & !rest_days_raw > 7)
head(incline_dumbbell_bulk)
```

```
##      date phase muscle_group      exercise weight_lbs set_number
## 1 2025-01-16 Bulk      Push Incline Dumbell Press      87.5         1
## 2 2025-01-16 Bulk      Push Incline Dumbell Press      87.5         2
## 3 2025-01-16 Bulk      Push Incline Dumbell Press      87.5         3
## 4 2025-02-06 Bulk      Push Incline Dumbell Press      87.5         2
## 5 2025-02-06 Bulk      Push Incline Dumbell Press      87.5         3
## 6 2025-02-10 Bulk      Push Incline Dumbell Press      90.0         1
##      reps bodyweight      notes volume one_rep_max start_of_week
## 1      8          NA Weight in Each hand 700.0    110.8333 2025-01-13
## 2      7          NA Weight in Each hand 612.5    107.9167 2025-01-13
## 3      6          NA Weight in Each hand 525.0    105.0000 2025-01-13
## 4      8          NA              700.0    110.8333 2025-02-03
## 5      5          NA              437.5    102.0833 2025-02-03
## 6      8          NA              720.0    114.0000 2025-02-10
##      rest_days_raw rest_days
## 1              3         3
## 2              0         0
## 3              0         0
## 4              0         0
## 5              0         0
## 6              2         2
```

```
dim(incline_dumbbell_bulk)
```

```
## [1] 19 14
```

```
#We lost 2 rows of data from removing NA
```

#I want to compare my strength in Incline Dumbbell Press by 1RM. If we choose to do volume, this can skew data due to the fact that the number of sets I chose to do could affect volume.

```
incline_dumbbell_bulk_model <- lm(one_rep_max ~ rest_days, data = incline_dumbbell_bulk)
incline_dumbbell_bulk_model
```

```
##
## Call:
## lm(formula = one_rep_max ~ rest_days, data = incline_dumbbell_bulk)
##
## Coefficients:
## (Intercept)      rest_days
##      111.126         1.018
```

```
summary(incline_dumbbell_bulk_model)
```

```
##
## Call:
## lm(formula = one_rep_max ~ rest_days, data = incline_dumbbell_bulk)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.0430 -3.1263 -0.1263  1.8553  9.2070
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   111.126     1.250   88.879  <2e-16 ***
## rest_days       1.018     1.162    0.876   0.393
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.754 on 17 degrees of freedom
## Multiple R-squared:  0.04323,    Adjusted R-squared:  -0.01305
## F-statistic: 0.7681 on 1 and 17 DF,  p-value: 0.393
```

```
#Slope: 1.69 -> Each additional rest day = +1.69 lbs on your Incline Press 1RM.
```

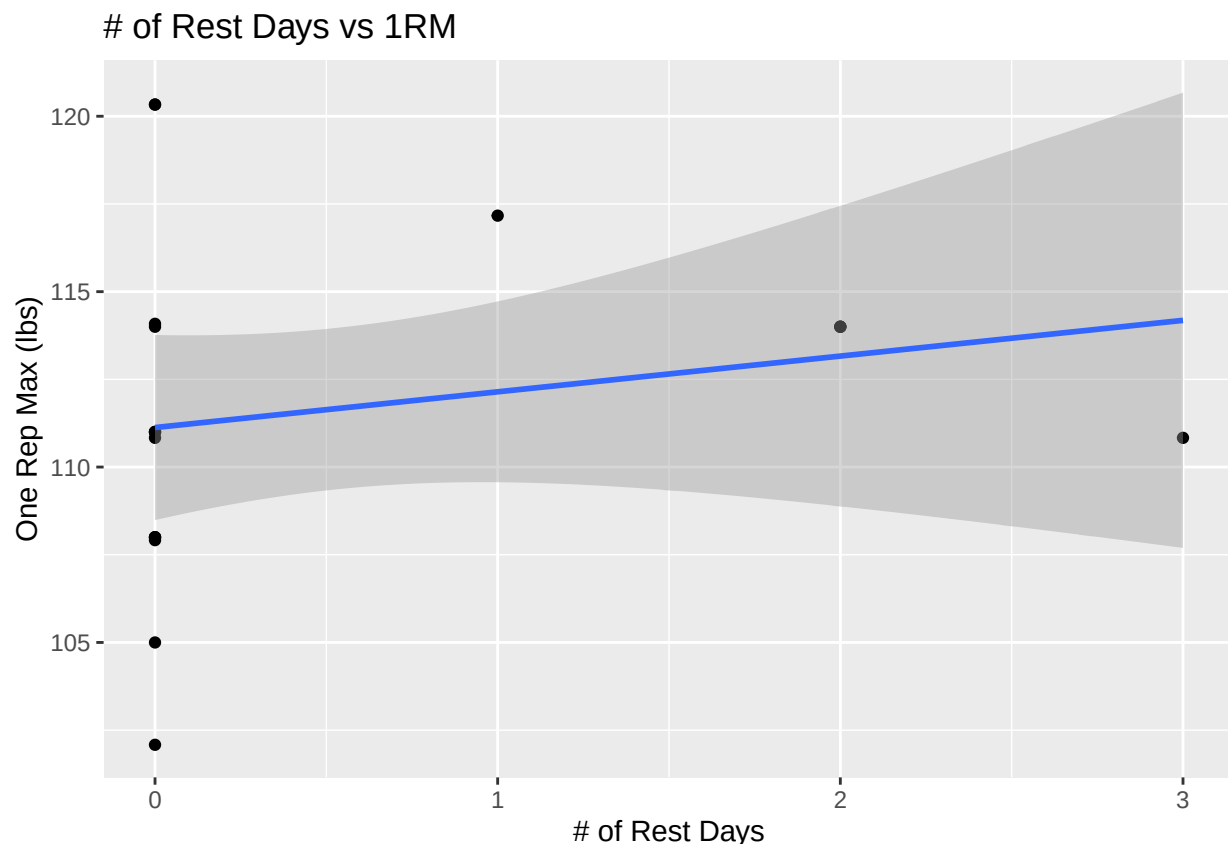
```
#p-value: 0.074 -> If rest truly had no effect (H0 was true), there is a 7.4% chance of seeing a slope
```

```
#The effect is SMALL! We can see that there is an average of 1.69 lbs per one rest day. However, there
```

#A linear regression was used to test whether rest days predicted Incline Dumbbell Press strength during the bulk. The model showed a positive trend (slope = 1.69 lbs per rest day), but the result was not statistically significant ($p = 0.074$, Adjusted $R^2 = 0.144$). This suggests that while rest may have had a small positive effect, the sample size was too small to draw a definitive conclusion. Future work with more data is needed to confirm this relationship.

```
#Make a regression line
ggplot(data = incline_dumbbell_bulk, aes(x = rest_days, y = one_rep_max)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "# of Rest Days vs 1RM",
    x = "# of Rest Days",
    y = "One Rep Max (lbs)"
  )
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



#Question 3: In terms of Bulking phase, let's analyze volume by muscle group and answer these questions:
 #a. Which muscle group did I train the most? (Most volume, sets) #b. How did my volume distribution change over time? (Weekly trends) #c. Which exercises drove the most volume for each muscle group
 #3.a. Which muscle group did I train the most? (Most volume, sets)

```
#We will make a new subset grouped by muscle group and create aggregates: total volume, total_sets(sets
volume_by_muscle_bulk <- bulk_hans_exercise_2025 %>%
  group_by(muscle_group) %>%
  summarise(
    total_volume = sum(volume, na.rm = TRUE),
    total_sets = n(),
    .groups = "drop"
  ) %>%
```

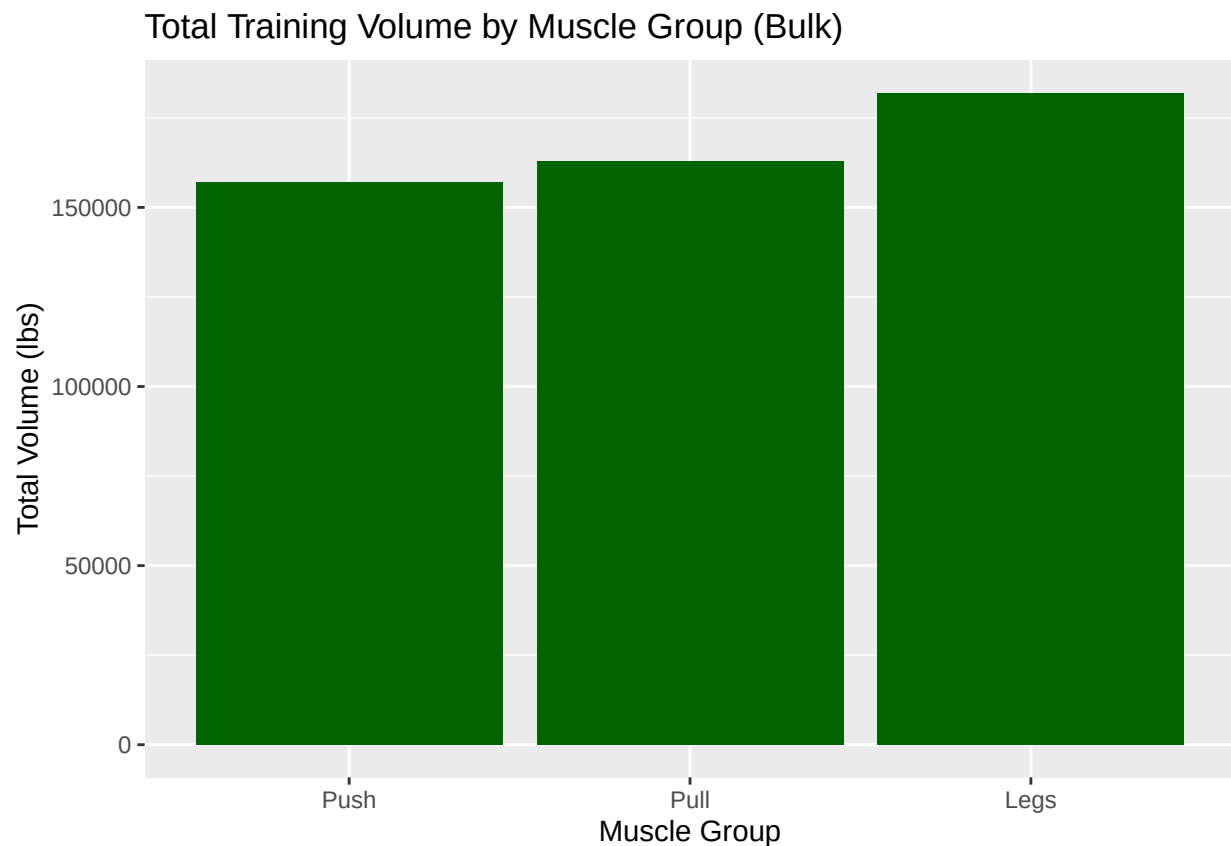
```
arrange(desc(total_volume))

volume_by_muscle_bulk
```

```
## # A tibble: 3 x 3
##   muscle_group total_volume total_sets
##   <chr>          <dbl>      <int>
## 1 Legs           181995         84
## 2 Pull           162844         194
## 3 Push           156948         231
```

```
#Visualize volume by muscle group subset
```

```
#We will use reorder() function on muscle_group column and order them by total volume
#geom_bar is not used here because that actually counts how many rows per muscle groups
#geom_col is used instead as it is used for aggregates (in this case, it's the total_volume we made)
ggplot(volume_by_muscle_bulk, aes(x = reorder(muscle_group, total_volume), y = total_volume)) +
  geom_col(fill = "darkgreen") +
  labs(
    title = "Total Training Volume by Muscle Group (Bulk)",
    x = "Muscle Group",
    y = "Total Volume (lbs)"
  )
```



```
#Conclusion 3. a.: The most volume was legs but this is due to how Leg exercises are naturally going to be
```

heavier per exercise. It's very interesting how Push is my favorite exercise but I did the least volume but had the most sets.

#3.b. How did my volume distribution change over time? (Weekly trends)

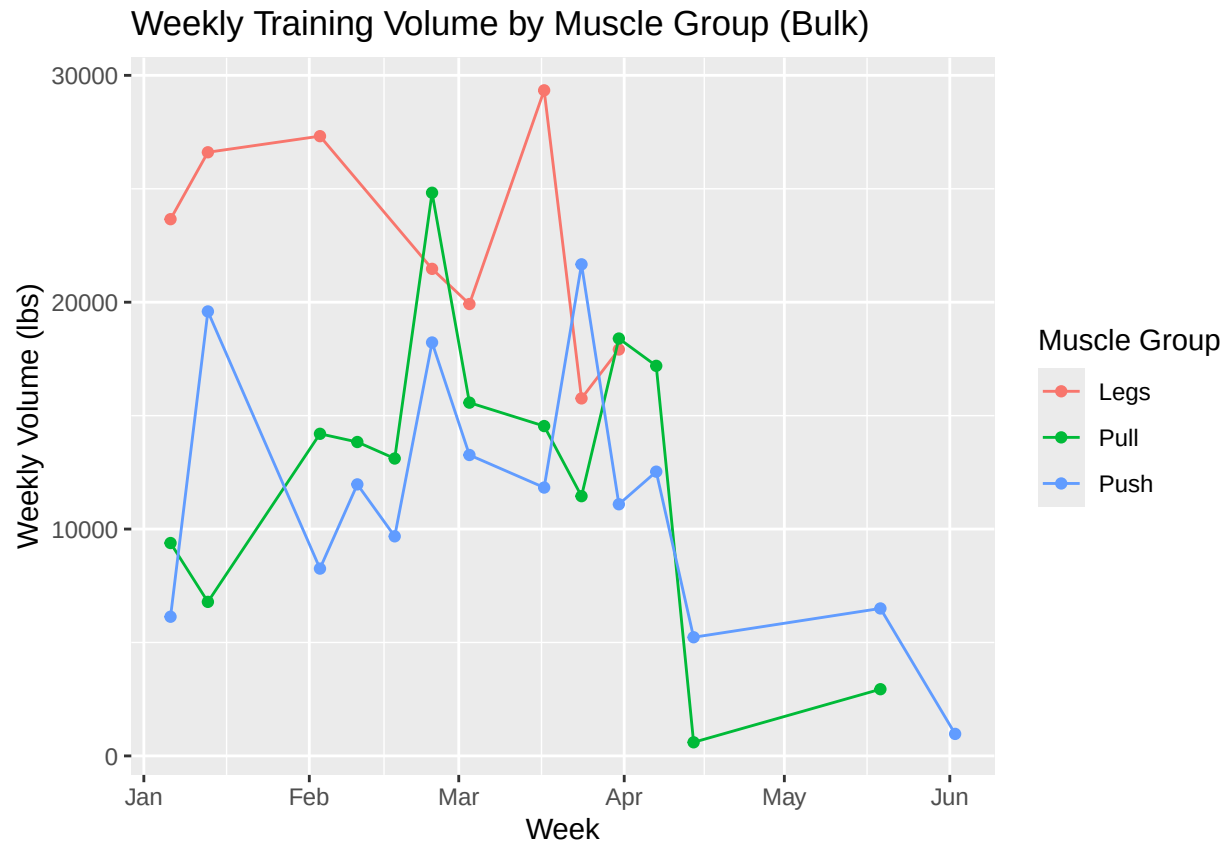
```
#We are making a subset that groups the bulking data by rows that belong to the same week and the muscle group
weekly_volume_muscle_bulk <- bulk_hans_exercise_2025 %>%
  group_by(start_of_week, muscle_group) %>%
  summarise(weekly_volume = sum(volume, na.rm = TRUE), .groups = "drop")

head(weekly_volume_muscle_bulk)
```

```
## # A tibble: 6 x 3
##   start_of_week muscle_group weekly_volume
##   <date>         <chr>         <dbl>
## 1 2025-01-06     Legs             23660
## 2 2025-01-06     Pull             9385
## 3 2025-01-06     Push             6135
## 4 2025-01-13     Legs             26610
## 5 2025-01-13     Pull             6790
## 6 2025-01-13     Push            19592.
```

#Visualize the above subset

```
#Make a legend which gives a distinct color per muscle group. Then we will add a scatterplot (geom_point) +
ggplot(weekly_volume_muscle_bulk, aes(x = start_of_week, y = weekly_volume, color = muscle_group)) +
  geom_point() +
  geom_line() +
  labs(
    title = "Weekly Training Volume by Muscle Group (Bulk)",
    x = "Week",
    y = "Weekly Volume (lbs)",
    color = "Muscle Group"
  )
```



#How come the volume decreases over time? Let's check if this is due to lack of data or just less volume

```
bulk_hans_exercise_2025 <- bulk_hans_exercise_2025 %>%
  mutate(start_of_month = floor_date(date, unit = "month")) #Add a column that converts every day (row)
```

```
bulk_monthly <- bulk_hans_exercise_2025 %>%
  group_by(start_of_month) %>%
  summarise(
    total_sets = n(),
    total_volume = sum(volume, na.rm = TRUE),
    .groups = "drop"
  )
```

```
bulk_monthly
```

```
## # A tibble: 6 x 3
##   start_of_month total_sets total_volume
##   <date>          <int>         <dbl>
## 1 2025-01-01         86         92172.
## 2 2025-02-01        176        162895
## 3 2025-03-01        163        164626.
## 4 2025-04-01         68         71682.
## 5 2025-05-01         14          9440
## 6 2025-06-01          2           970
```

#Conclusion 3.b.: While weekly volume fluctuated throughout my bulk, it can be observed that as I was about to enter my cut, the overall volume decreased for all muscle groups. This is prominent approximately

during the month of April. This is also when I stopped doing Legs. I personally don't like hitting legs so this is consistent with my training regime. I also looked back at the raw data from my notes and found "WE BACKK(havent been tracking but still going to failure)". So this was likely due to a combination of scarce data logging and going into my cut. I stopped going to the gym seriously during this time due to exams and busy with school and work. It is interesting to see how I did 13 more sets in February compared to March, yet I was able to do more volume

#3. c. Which exercises drove the most volume for each muscle group

```
bulk_exercise <- bulk_hans_exercise_2025 %>%
  group_by(exercise) %>%
  summarise(
    total_sets = n(),
    total_volume = sum(volume, na.rm = TRUE),
    .groups = "drop") %>%
  arrange(desc(total_sets))

head(bulk_exercise)
```

```
## # A tibble: 6 x 3
##   exercise                total_sets total_volume
##   <chr>                  <int>      <dbl>
## 1 "Cable Lateral Raise"      36      10835
## 2 "Standing Dumbbell Curls"  34      10359.
## 3 "Rope Overhead Tricep"    30      22080
## 4 "Rope Pushdown Tricep"    28      20110
## 5 "Incline Dumbbell Press "  22      13888.
## 6 "Leg Extensions"         21      46195
```

```
#We have simply too many exercises so we will simplify this to our top 10 exercises
top_exercises_bulk_volume <- bulk_exercise %>%
  slice_max(total_volume, n = 10) %>% #slicing trims the data and gives us only n amount
  pull(exercise) #Pull works similar to $ where it gives us the column values as a vector

top_exercises_bulk_volume
```

```
## [1] "Leg Extensions"          "Hack Squats"
## [3] "Standing Calf Raises"    "Leg Press"
## [5] "Seated Leg Curls"        "Rope Hammer Curls"
## [7] "Rope Overhead Tricep"    "Rope Pushdown Tricep"
## [9] "Smith Machine Incline Press" "Seated Cable Chest Press"
```

```
library(tidyr)
```

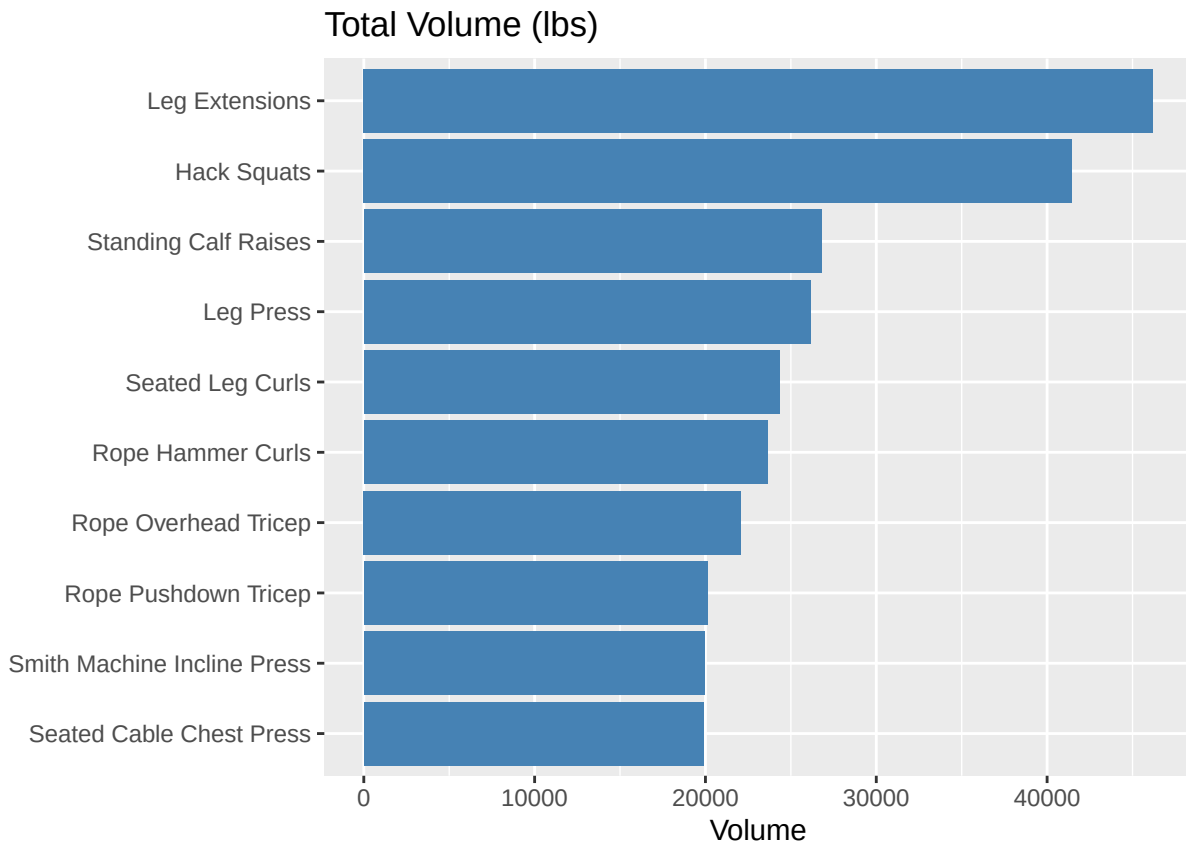
```
## Warning: package 'tidyr' was built under R version 4.5.3
```

```
#Let's make a subset so we can compare the exercises by volume
```

```
#reorder(x-axis, sort by "this" data)
bulk_by_volume_plot <- bulk_exercise %>%
  filter(exercise %in% top_exercises_bulk_volume) %>%
  ggplot(aes(x = reorder(exercise, total_volume), y = total_volume)) + #sort x-axis by exercise name and
```

```
geom_col(fill = "steelblue") +
coord_flip() + #flips graph so we can see the names and they don't mush up
labs(title = "Total Volume (lbs)", x = "", y = "Volume")
```

bulk_by_volume_plot



#Do the same thing but for sets: Let's make a subset so we can compare the exercises by sets

```
top_exercises_bulk_sets <- bulk_exercise %>%
  slice_max(total_sets, n = 10) %>%
  pull(exercise) #Pull works similar to $ where it gives us the column values as a vector
```

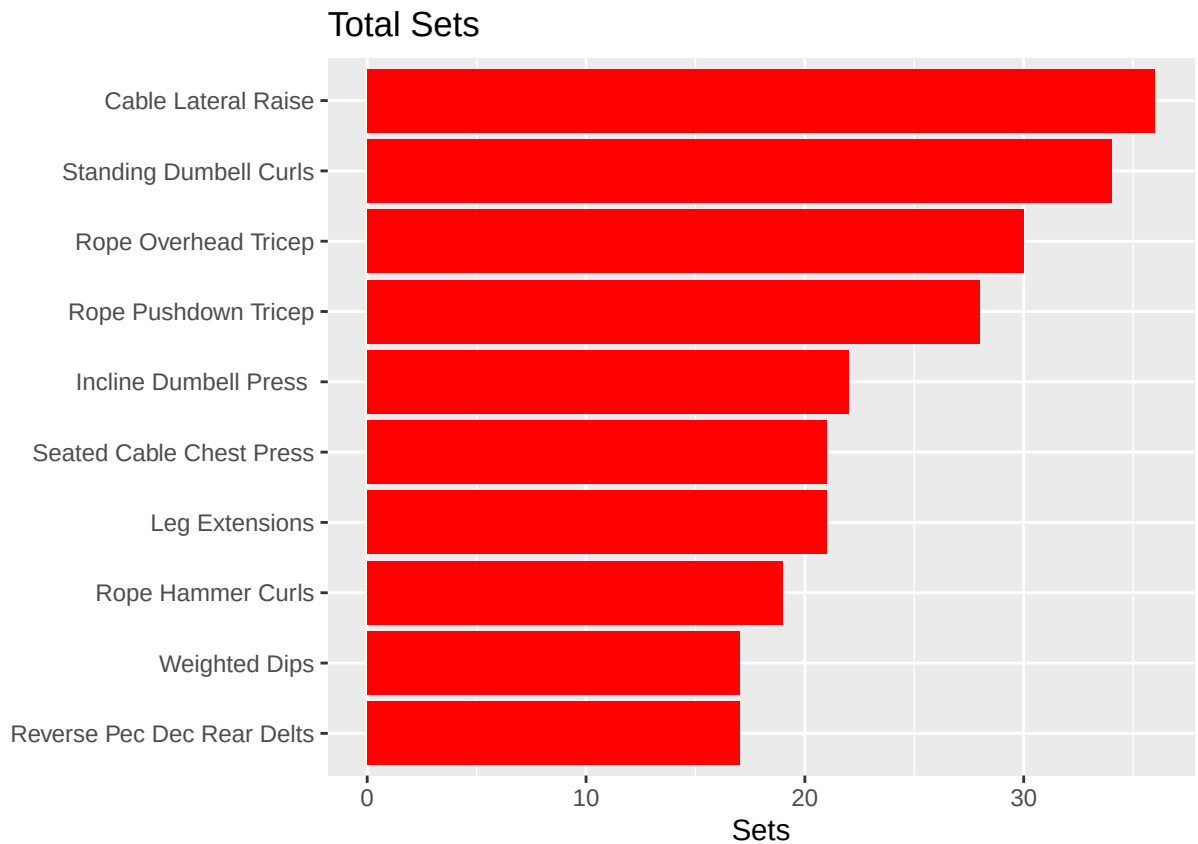
top_exercises_bulk_sets

```
## [1] "Cable Lateral Raise"      "Standing Dumbbell Curls"
## [3] "Rope Overhead Tricep"    "Rope Pushdown Tricep"
## [5] "Incline Dumbbell Press "  "Leg Extensions"
## [7] "Seated Cable Chest Press" "Rope Hammer Curls"
## [9] "Reverse Pec Dec Rear Delts" "Weighted Dips"
```

```
bulk_by_sets_plot <- bulk_exercise %>%
  filter(exercise %in% top_exercises_bulk_sets) %>%
  ggplot(aes(x = reorder(exercise, total_sets), y = total_sets)) +
  geom_col(fill = "red") +
  coord_flip() +
```

```
labs(title = "Total Sets", x = "", y = "Sets")

bulk_by_sets_plot
```



#Conclusion 3.c.: In terms of volume, it is to no surprise that Leg workouts naturally make up the top 5 by total volume. This is because Leg workouts can usually be done at a heavier weight while keeping the rep range similar to upper body exercises. However, It is interesting to see that Push workouts dominate in terms of total sets done. This is likely due to how earlier, we saw that I stopped doing legs towards the end of my bulk. It is still surprising however that leg extensions are still in the top 10 for total sets.

#Question 4: Do the same analysis but for the cutting phase. Additionally I want to visualize my bodyweight
 #4.a. Total volume by Muscle Group (cut)

```
volume_by_muscle_cut <- cut_hans_exercise_2025 %>%
  group_by(muscle_group) %>%
  summarise(
    total_volume = sum(volume, na.rm = TRUE),
    total_sets = n(),
    .groups = "drop"
  ) %>%
  arrange(desc(total_volume))

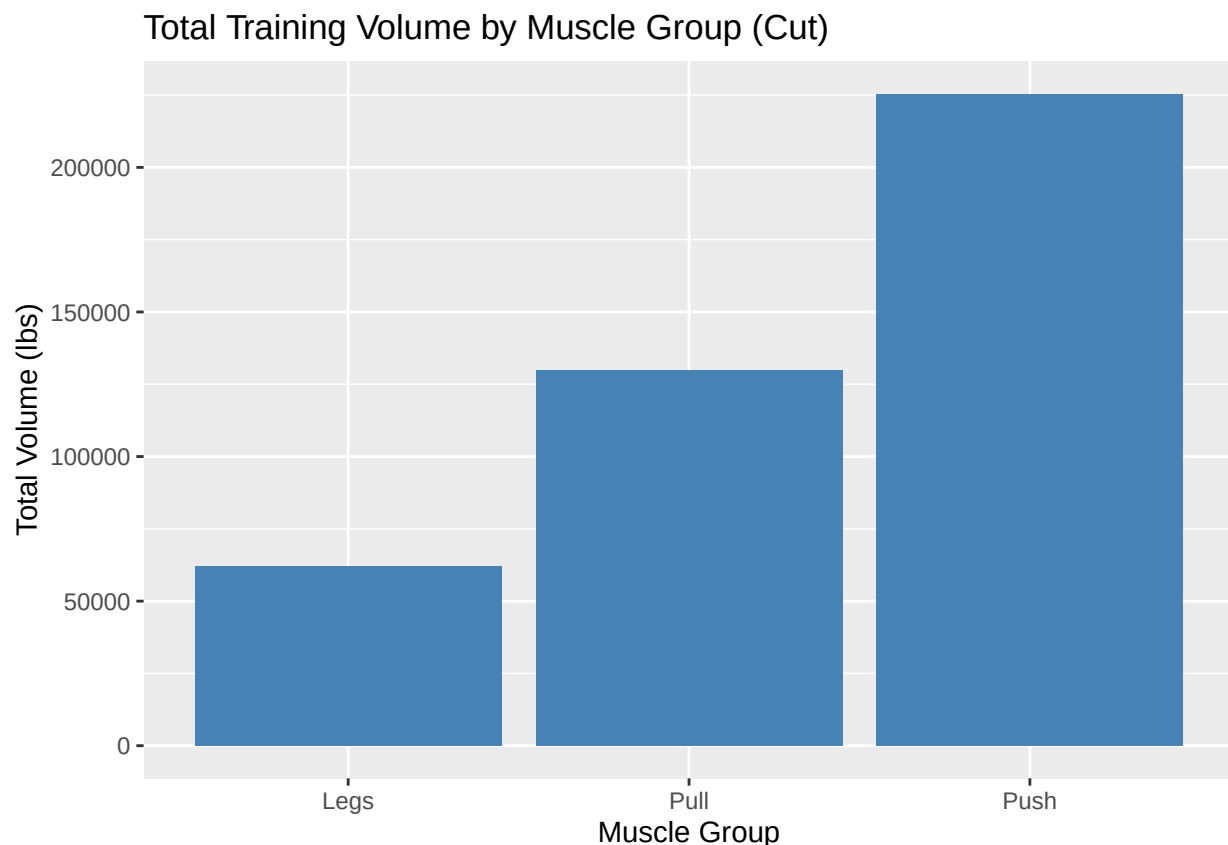
# View the table
volume_by_muscle_cut
```

```
## # A tibble: 3 x 3
```

```
## muscle_group total_volume total_sets
## <chr>          <dbl>      <int>
## 1 Push          225470       337
## 2 Pull          129755       186
## 3 Legs           61955        31
```

#Visualize above subset

```
ggplot(volume_by_muscle_cut, aes(x = reorder(muscle_group, total_volume), y = total_volume)) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Total Training Volume by Muscle Group (Cut)",
    x = "Muscle Group",
    y = "Total Volume (lbs)"
  )
```



#Conclusion 4.a: When comparing total volume for legs in my cut compared to my bulk, it can be observed that it is the complete opposite of what happened during my bulk. I did a total of 554 sets during my cut so I exercised more often compared to 512 sets in total during my bulk. Similarly, I did push the most (Cut: Volume of 225470 (337 sets) vs. Bulk: Volume of 154687.5 (228 sets)). However, when we look at legs, I only did 31 sets in total during my cut compared to the 84 I did in my bulk. Then for pull, it ranked in the middle with a similar set count but less volume during my cut (165103.8 in bulk and 129755 in cut). Overall, it can be seen that although I worked out more often, the concentration of these workouts in terms of muscle group were different.

#4.b.

```

weekly_volume_muscle_cut <- cut_hans_exercise_2025 %>%
  group_by(start_of_week, muscle_group) %>%
  summarise(
    weekly_volume = sum(volume, na.rm = TRUE),
    .groups = "drop")

head(weekly_volume_muscle_cut)

```

```

## # A tibble: 6 x 3
##   start_of_week muscle_group weekly_volume
##   <date>         <chr>         <dbl>
## 1 2025-06-09     Pull             7760
## 2 2025-06-09     Push            20672.
## 3 2025-06-16     Legs            25500
## 4 2025-06-16     Pull            19335
## 5 2025-06-16     Push             4080
## 6 2025-06-23     Legs            21525

```

```

#I added method = "lm" to see the line of best fit
ggplot(weekly_volume_muscle_cut, aes(x = start_of_week, y = weekly_volume, color = muscle_group)) +
  geom_line() +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(
    title = "Weekly Training Volume by Muscle Group (Cut)",
    x = "Week",
    y = "Weekly Volume (lbs)",
    color = "Muscle Group"
  )

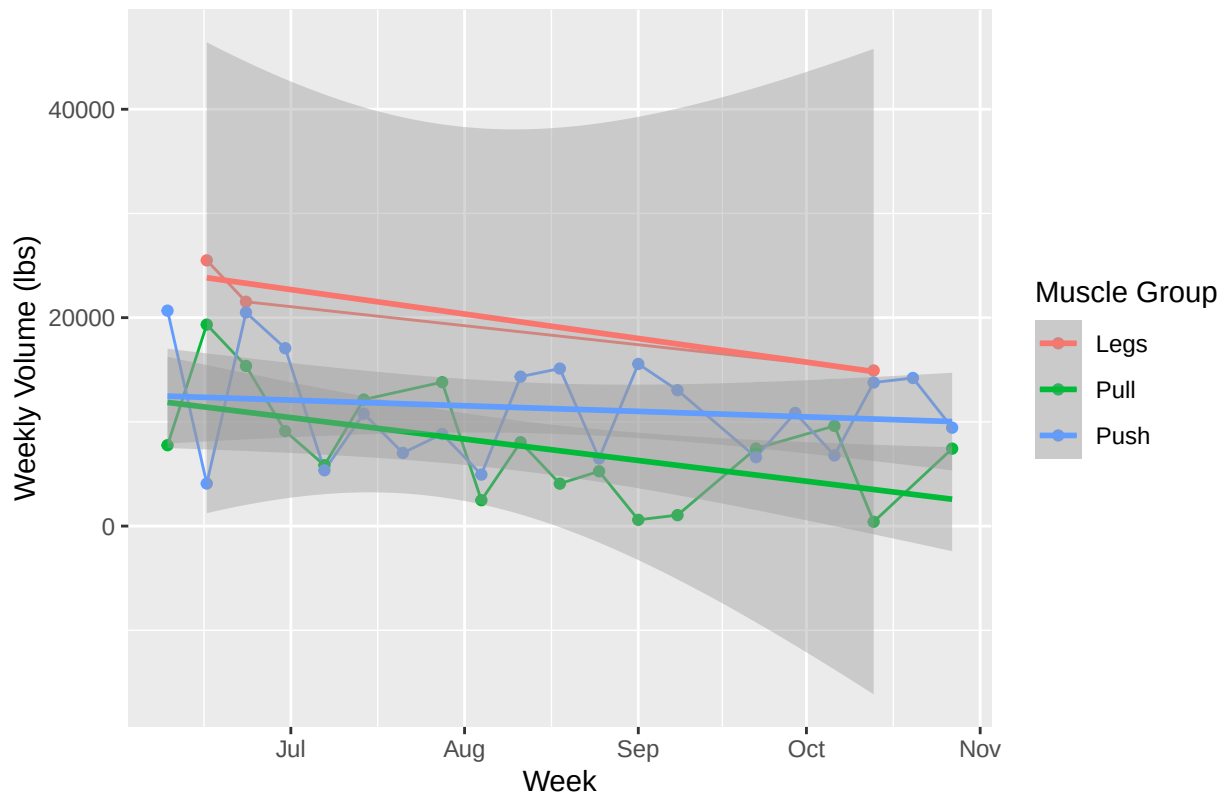
```

```

## 'geom_smooth()' using formula = 'y ~ x'

```

Weekly Training Volume by Muscle Group (Cut)



#Conclusion 4.b.: To no surprise during my cut, there were only 3 weeks of Leg exercises and a huge gap between July to October. When looking at the trend for Push and Pull, I noticed that as I approached the end of my cut, the volume stayed the same with non-negotiable slight decrease. This is honestly what I was hoping to see in my data because I was hoping to not lose any strength in my cut. To no surprise, I lost a minimal amount of strength and I felt great during my workouts without any feeling of lack of energy.

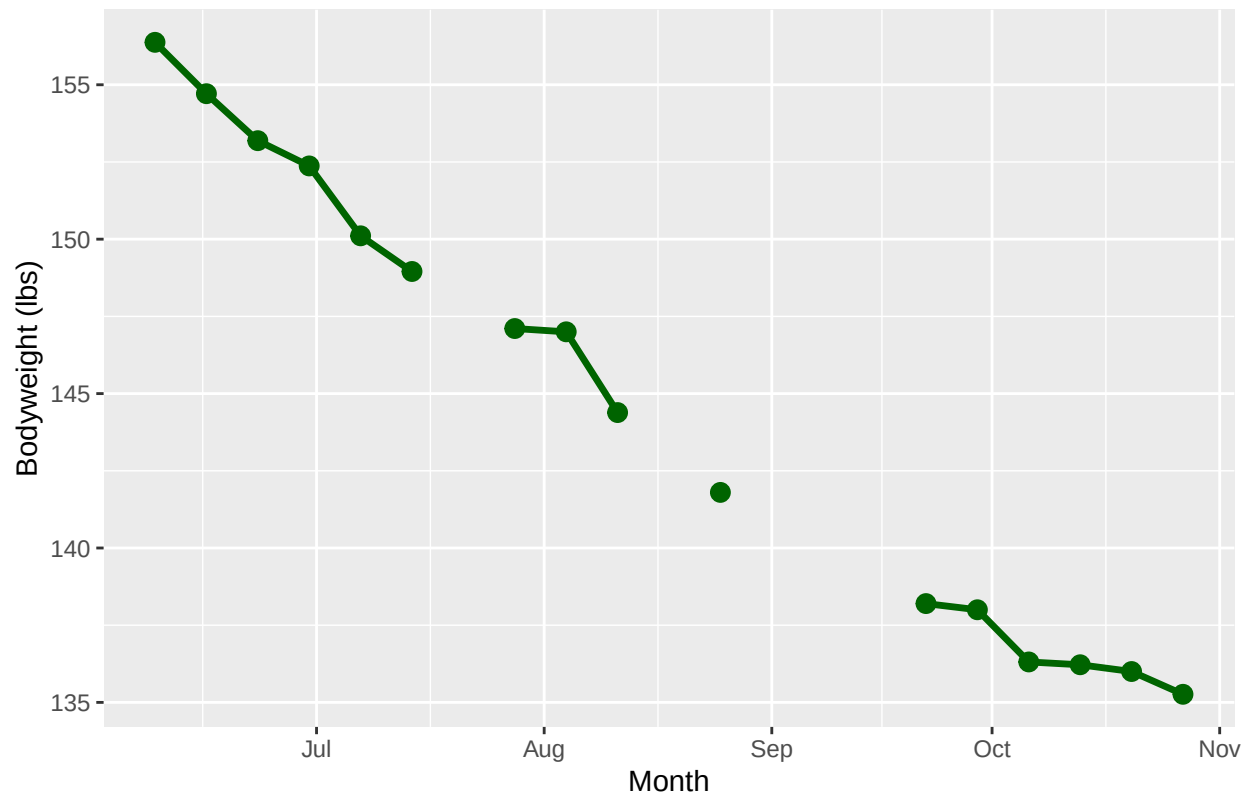
#Now I want to visualize the cutting subset in terms of bodyweight

```
#This plot will have gaps due to the fact that there is missing data that was not logged.
ggplot(cut_weekly_total, aes(x = start_of_week, y = avg_bodyweight)) +
  geom_line(color = "darkgreen", size = 1.2) +
  geom_point(color = "darkgreen", size = 3) +
  labs(
    title = "Weekly Average Bodyweight During Cut",
    x = "Month",
    y = "Bodyweight (lbs)"
  )
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Weekly Average Bodyweight During Cut



#Looking at the data, there is a strong linear decreasing relationship between bodyweight and time.

#Now let's analyze the full data (cut and bulk) and attempt to analyze any trends and patterns within the data.

#Make total weekly volume for each phase but for full data

#Again, add a new column to original data which rounds down to the start of the week

```
hans_exercise_2025 <- hans_exercise_2025 %>%
  mutate(start_of_week = floor_date(date, unit = "week", week_start = 1))
```

```
full_weekly_volume <- hans_exercise_2025 %>%
  group_by(start_of_week, phase) %>%
  summarise(total_volume = sum(volume, na.rm = TRUE),
    .groups = "drop")
```

```
head(full_weekly_volume)
```

```
## # A tibble: 6 x 3
##   start_of_week phase total_volume
##   <date>         <chr>         <dbl>
## 1 2025-01-06     Bulk           39180
## 2 2025-01-13     Bulk           52992.
## 3 2025-02-03     Bulk           49778.
## 4 2025-02-10     Bulk           25808.
## 5 2025-02-17     Bulk            22790
## 6 2025-02-24     Bulk           64520
```

#Visualize the above subset to observe trends in volume per week

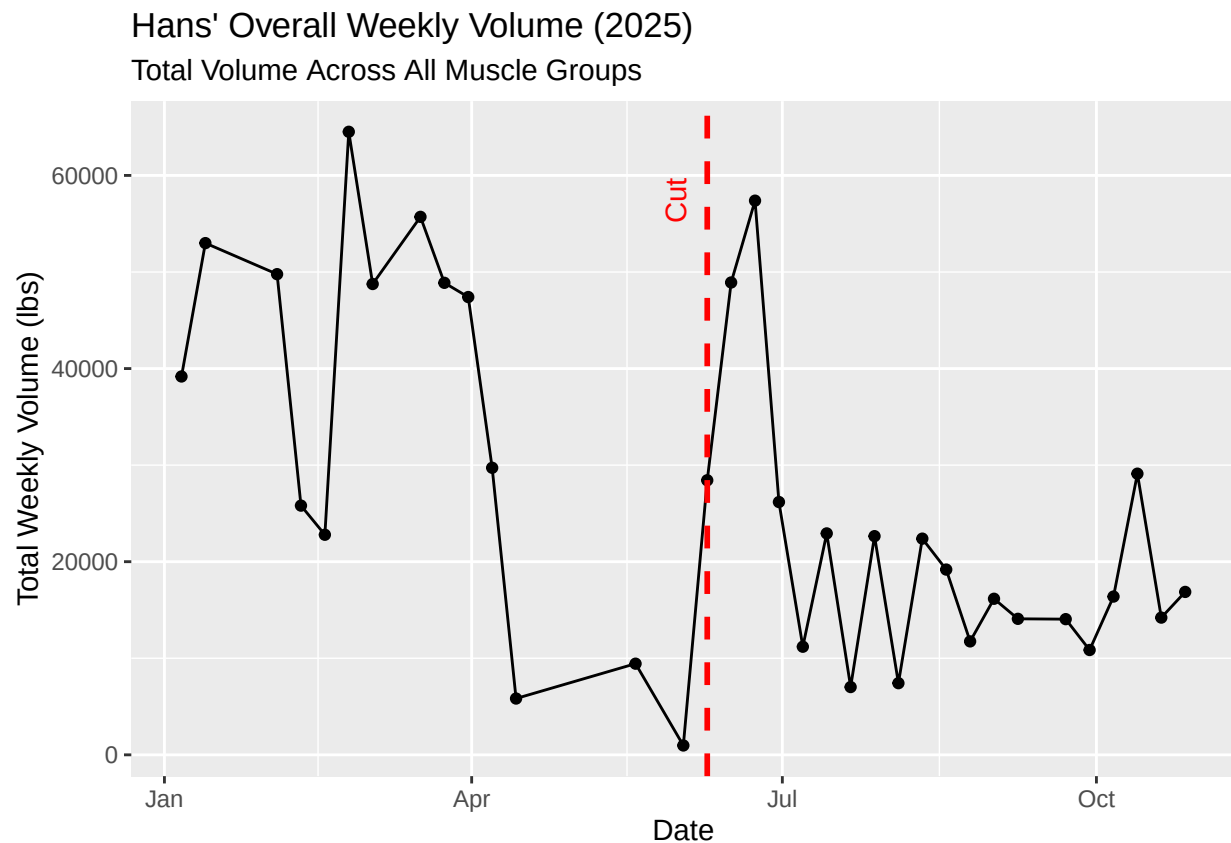
#Create a visualization

#This is going to include a scatterplot that are connected by lines

#It is also going to have a vline (vertical line)

#We are going to annotate a text onto the visualization. I use la annotate() to mark the start of Cut's

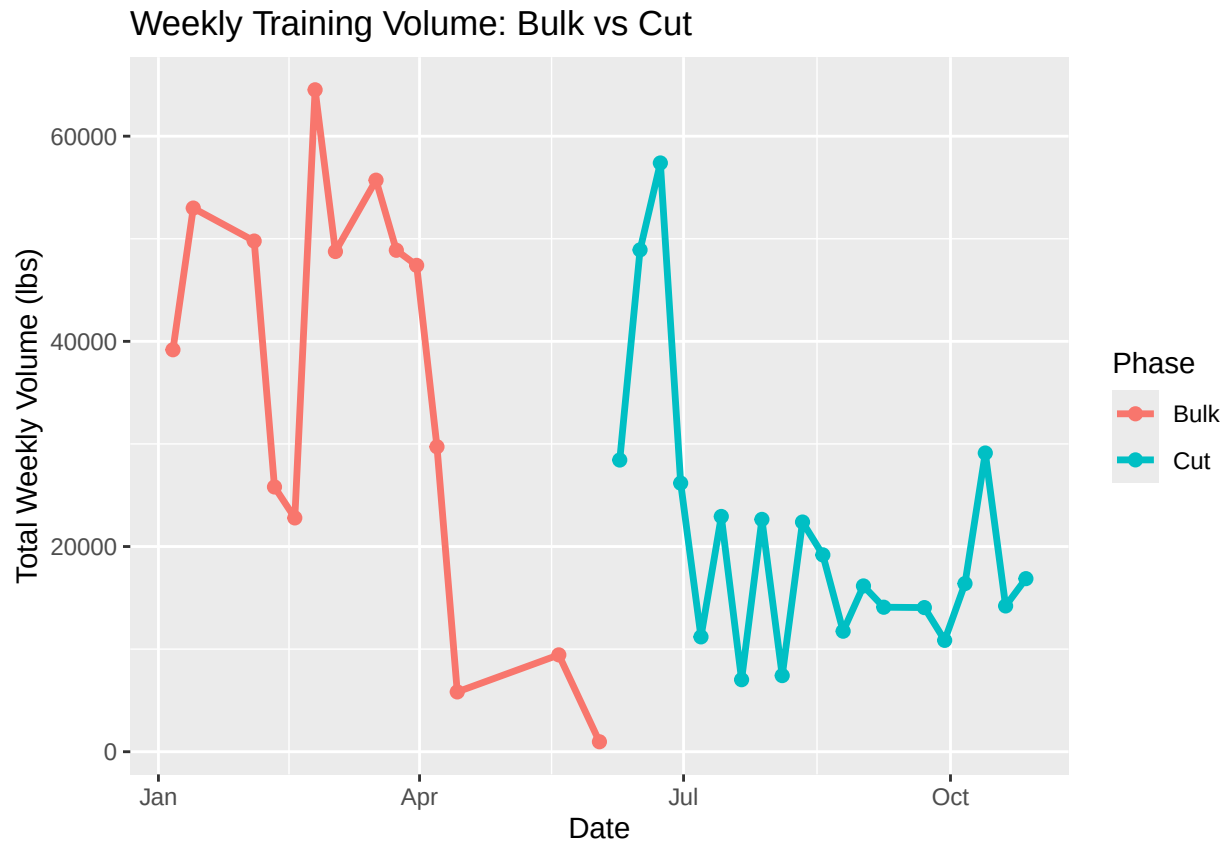
```
ggplot(full_weekly_volume, aes(x = start_of_week, y = total_volume)) +  
  geom_point() +  
  geom_line() +  
  geom_vline(xintercept = as.Date("2025-06-09"), linetype = "dashed", color = "red", size = 1) +  
  annotate("text", x = as.Date("2025-06-09"), y = max(full_weekly_volume$total_volume, na.rm = TRUE), label =  
    "Cut",  
    x = "Date",  
    y = "Total Weekly Volume (lbs)"  
  )
```



#Visualize this again but we will plot the Volume by Phase (similar graph)

```
ggplot(full_weekly_volume, aes(x = start_of_week, y = total_volume, color = phase)) +  
  geom_line(size = 1.2) +  
  geom_point(size = 2) +  
  labs(  
    title = "Weekly Training Volume: Bulk vs Cut",  
    x = "Date",  
    y = "Total Weekly Volume (lbs)",  
    color = "Phase"
```

)



#Observation: The first thing that was prominent in the data was how much more consistent I was during the cut. I notice a lot more data points within this Phase. Naturally, I notice that there is not that much variance in my volume. I am happy to see this as it partially signifies that I did not lose that much strength and did higher reps even though I did less weight over time. However, it is obvious that although that data is lacking in the Bulk Phase, the data point peaks were higher compared to my cut. This is clear that I did higher reps and weight lifted in my bulk.

#Now we will analyze Volume by Muscle Group

```
full_weekly_muscle <- hans_exercise_2025 %>%
  group_by(start_of_week, muscle_group) %>%
  summarise(total_volume = sum(volume, na.rm = TRUE),
            .groups = "drop")

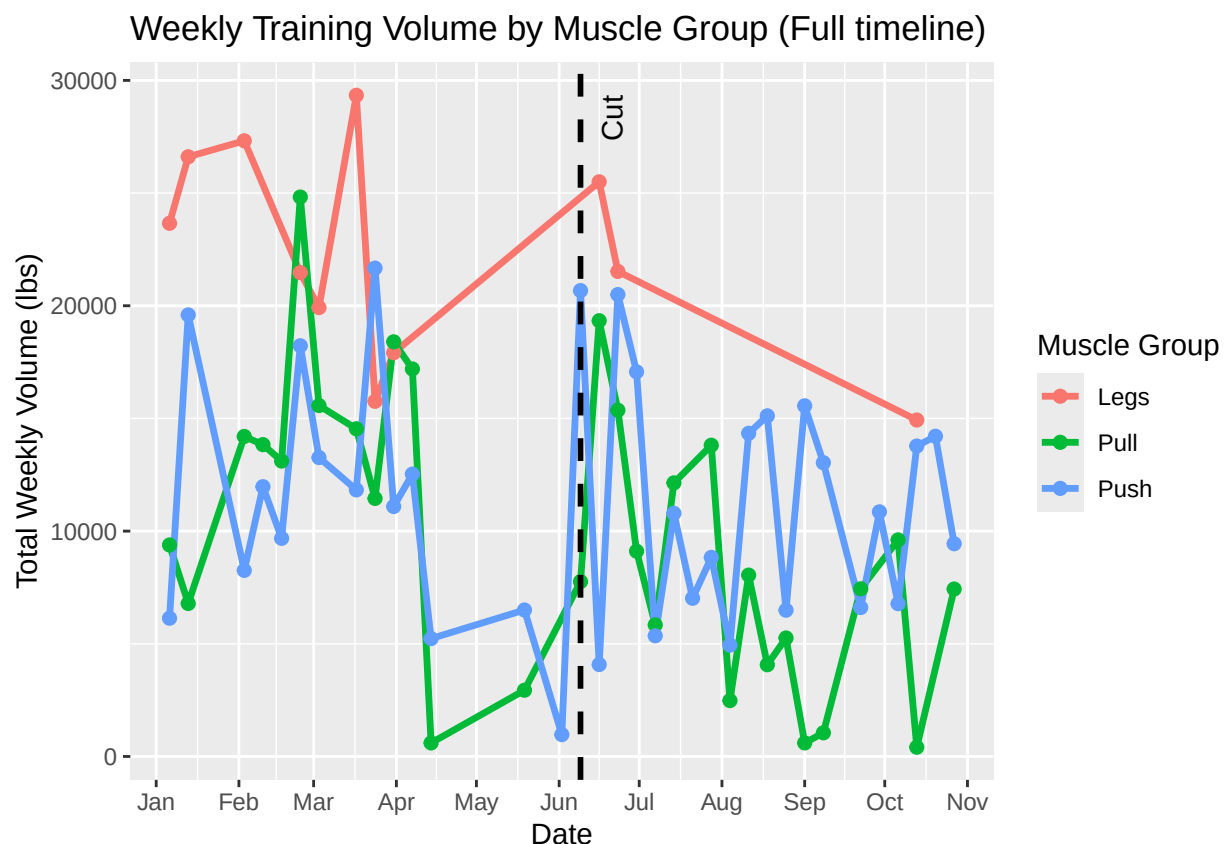
head(full_weekly_muscle)
```

```
## # A tibble: 6 x 3
##   start_of_week muscle_group total_volume
##   <date>         <chr>         <dbl>
## 1 2025-01-06     Legs           23660
## 2 2025-01-06     Pull           9385
## 3 2025-01-06     Push           6135
## 4 2025-01-13     Legs          26610
## 5 2025-01-13     Pull           6790
```

```
## 6 2025-01-13    Push    19592.
```

```
#Visualize subset
```

```
ggplot(full_weekly_muscle, aes(x = start_of_week, y = total_volume, color = muscle_group)) +  
  geom_line(size = 1.2) +  
  geom_point(size = 2) +  
  geom_vline(xintercept = as.Date("2025-06-09"), linetype = "dashed", color = "black", size = 1) +  
  annotate("text", x = as.Date("2025-06-09"), y = max(full_weekly_muscle$total_volume, na.rm = TRUE), label = "Cut") +  
  scale_x_date(  
    date_breaks = "1 month", # Show every month  
    date_labels = "%b"       # Format: "Jan", "Feb"  
  ) +  
  labs(  
    title = "Weekly Training Volume by Muscle Group (Full timeline)",  
    x = "Date",  
    y = "Total Weekly Volume (lbs)",  
    color = "Muscle Group"  
  )
```



#Observation: When looking at the volume by muscle group data, again the legs consist of the least amount of data points. It also looks like the time ranging from Apr-Jul have the least amount of data points which is when I was transitioning into my cut. This signifies that as I was transitioning, I just did not log as much data as I could and that this was a busier time in my life due to exams in April. During my bulk, there seems to have been a honeymoon phase from Feb-Apr for all my muscle groups. This must be when I was really focused on the gym and making sure that I was doing the bulk properly, making sure that my strength was

proportional to me gaining weight (although I had not logged my weight during this phase). All my muscle groups have a recurring theme in which the volume naturally progressed as time went on during my cut.